

Mycelic: where the hidden patterns go

Loss accounting, gap decomposition and the vNext results (interim, before the final rerun)

QUESTION

How should thousands of private user-level agents abstract, route, combine, question, verify and propagate what they know upward, so that an enterprise kernel discovers what no individual could?

ENTERPRISE SIZES MEASURED

— users

CONTENTS

9 sections · 40 tables · 0 figures

GENERATED

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Every table and figure in this document is computed from the raw per-run records in `research/mycelic/artifacts/` at generation time. No number is transcribed by hand. Section 5 states exactly which results were measured on real models and which were simulated.

What this is. The v1 benchmark said the hierarchy found 34% of the hidden patterns at 50,000 users and the strongest centralised system found 69%. This document says *where the other patterns went*. Every discoverable gold pattern is traced through the pipeline and the first stage at which it is lost is recorded, for the hierarchy (`H_mycelic_full`), its centralised twin (`B4_central_triage`), the perfect-retrieval oracle (`Y_oracle_retrieval`) and the strongest centralised baseline (`A2_chunked_ctx`), on the same worlds. Then three paired diagnostics on identical worlds decompose the gap into the stages responsible.

Nothing here is a design. It is the measurement the next design is built on. Every number is computed from

`research/mycelic/artifacts/loss_funnel.jsonl` and `quick_diag*.jsonl` at generation time. 5 seeds at 10,000 users, 3 at 50,000.

1. The finding in one paragraph

The handoff pack's leading hypothesis was that the sketch channel is too poor to seed good hypotheses. It is not: at 10,000 users the sketch triage sees 98.5% of gold anchors. The pattern dies later, at two places. First, the **question budget**: the calibrated budget asks only 63.5% of gold anchors at 10k and 43% at 50k, and that is the single largest first-loss stage at both scales (47% and 51% of the gap to A2). Second, the **register**: at 10k, 12% of all patterns are matched at confidence ≥ 0.5 and then cut, because the hierarchy rates spurious candidates as highly as genuine ones and the register holds one entry per entity. A2 loses nothing at the register. Asking every triage candidate recovers evidence for 98.5% of patterns and the kernel forms a correct confident candidate for ~80% — more than A2 — and then buries them. The calibrated small budget was the right choice *given the ranker*; the ranker is the fault. At 50,000 users the sketch additionally loses 18% of patterns (mostly rare: a single-witness facet cannot set a site bit under the support threshold), so scale adds a third, smaller stage.

2. Stage-by-stage survival

Stages are ordered by pipeline position. They are not strictly nested — a pattern can be visible to the sketch without having been extracted — so `first stage lost` records the first stage in order that failed. The flat systems have no sketch, triage, question or descent stage.

10,000 users (5 seeds)

Stage survival, all patterns

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (≥ 2 facets, right entity+predicate)	—	0.960 [0.93, 0.99]	0.960 [0.93, 0.99]	0.960 [0.93, 0.99]
visible to sketch triage (≥ 2 foreign sites, ≥ 2 regions, span ≥ 2)	—	—	0.985 [0.97, 0.99]	—
on the triage candidate list	—	—	0.985 [0.97, 0.99]	—
inside the question budget	—	—	0.635 [0.58, 0.70]	—
descent reached a facet holder	—	—	0.665 [0.61, 0.73]	—
≥ 2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.970 [0.95, 0.99]	0.690 [0.64, 0.74]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.885 [0.84, 0.92]	0.870 [0.82, 0.92]	0.595 [0.54, 0.65]	0.865 [0.85, 0.89]
matched primary rule at ANY confidence	0.855 [0.80, 0.89]	0.770 [0.74, 0.81]	0.540 [0.45, 0.63]	0.770 [0.72, 0.82]
matched with confidence ≥ 0.5	0.810 [0.74, 0.86]	0.760 [0.73, 0.79]	0.515 [0.44, 0.61]	0.745 [0.70, 0.79]
survived the register cut (= reported)	0.685 [0.62, 0.74]	0.370 [0.32, 0.42]	0.375 [0.32, 0.43]	0.240 [0.19, 0.28]

patterns counted: A2_chunked_ctx: 200, B4_central_triage: 200, H_mycelic_full: 200, Y_oracle_retrieval: 200

Stage survival, rare-signal patterns only

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (≥ 2 facets, right entity+predicate)	—	0.910 [0.83, 0.99]	0.910 [0.83, 0.99]	0.910 [0.83, 0.99]
visible to sketch triage (≥ 2 foreign sites, ≥ 2 regions, span ≥ 2)	—	—	0.977 [0.95, 1.00]	—

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
on the triage candidate list	—	—	0.977 [0.95, 1.00]	—
inside the question budget	—	—	0.446 [0.36, 0.54]	—
descent reached a facet holder	—	—	0.463 [0.41, 0.53]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.955 [0.92, 0.99]	0.500 [0.42, 0.58]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.875 [0.82, 0.92]	0.801 [0.76, 0.86]	0.416 [0.30, 0.54]	0.853 [0.82, 0.89]
matched primary rule at ANY confidence	0.851 [0.75, 0.92]	0.703 [0.65, 0.77]	0.360 [0.26, 0.46]	0.772 [0.69, 0.85]
matched with confidence >= 0.5	0.773 [0.62, 0.90]	0.703 [0.65, 0.77]	0.341 [0.25, 0.45]	0.737 [0.67, 0.80]
survived the register cut (= reported)	0.653 [0.55, 0.75]	0.258 [0.19, 0.33]	0.215 [0.16, 0.27]	0.143 [0.10, 0.18]

patterns counted: A2_chunked_ctx: 78, B4_central_triage: 78, H_mycelic_full: 78, Y_oracle_retrieval: 78

Stage survival, common patterns only

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (>=2 facets, right entity+predicate)	—	1.000 [1.00, 1.00]	1.000 [1.00, 1.00]	1.000 [1.00, 1.00]
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	—	—	0.993 [0.98, 1.00]	—
on the triage candidate list	—	—	0.993 [0.98, 1.00]	—
inside the question budget	—	—	0.771 [0.66, 0.85]	—
descent reached a facet holder	—	—	0.798 [0.72, 0.88]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.986 [0.96, 1.00]	0.821 [0.76, 0.88]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.893 [0.86, 0.93]	0.916 [0.86, 0.98]	0.721 [0.69, 0.76]	0.870 [0.84, 0.90]
matched primary rule at ANY confidence	0.860 [0.84, 0.89]	0.818 [0.79, 0.85]	0.660 [0.56, 0.75]	0.761 [0.72, 0.81]
matched with confidence >= 0.5	0.845 [0.83, 0.86]	0.804 [0.77, 0.84]	0.630 [0.54, 0.72]	0.746 [0.70, 0.79]
survived the register cut (= reported)	0.716 [0.67, 0.76]	0.450 [0.34, 0.55]	0.484 [0.43, 0.55]	0.307 [0.25, 0.35]

patterns counted: A2_chunked_ctx: 122, B4_central_triage: 122, H_mycelic_full: 122, Y_oracle_retrieval: 122

Where the hierarchy loses each pattern (terminal stage)

stage the pattern died at	all	rare	common
extracted (>=2 facets, right entity+predicate)	0 (0%)	0	0
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	3 (2%)	2	1
on the triage candidate list	0 (0%)	0	0
inside the question budget	58 (29%)	37	21
descent reached a facet holder	0 (0%)	0	0
>=2 gold links in kernel pool (= evidence coverage)	0 (0%)	0	0
candidate formed on right entity + chain	17 (8%)	7	10
matched primary rule at ANY confidence	14 (7%)	5	9
matched, but confidence < 0.5	5 (2%)	1	4
cut from the register (matched at >= 0.5, out-ranked)	28 (14%)	10	18
reported	75 (38%)	16	59
total	200	78	122

Why the sketch could not see the invisible ones

why the sketch could not see it	all	rare	common	median witnesses
causal span < 2 across foreign sites	3	2	1	7

Discovery by number of witnesses (hierarchy)

witnesses (records)	n	visible to sketch	in pool	reported
0–4	12	1.00	0.75	0.25
5–7	52	0.96	0.46	0.21
8–11	16	1.00	0.44	0.12
12–16	13	1.00	0.62	0.15
17–25	65	0.98	0.78	0.45
26–∞	42	1.00	0.93	0.67

Discovery by number of witnesses (A2_chunked_ctx)

witnesses (records)	n	visible to sketch	in pool	reported
0–4	12	—	1.00	0.67
5–7	52	—	1.00	0.56
8–11	16	—	1.00	0.81
12–16	13	—	1.00	0.77
17–25	65	—	1.00	0.74
26–∞	42	—	1.00	0.69

Where the reported patterns sit in the register

architecture	reported	median rank (0-based, confidence-sorted register)	within top 40	within top 100	median confidence
A2_chunked_ctx	137	197	19	37	0.85
B4_central_triage	74	263	9	16	0.88
H_mycelic_full	75	255	8	18	0.89
Y_oracle_retrieval	48	279	3	9	0.91

Decomposing the gap to the strongest centralised system

Paired on 200 (seed, pattern) pairs at 10,000 users: A2_chunked_ctx reports 137 (68.5%), H_mycelic_full reports 75 (37.5%). A2_chunked_ctx finds 77 that H_mycelic_full misses; H_mycelic_full finds 15 that A2_chunked_ctx misses.

Of the 77 patterns A2_chunked_ctx reports and H_mycelic_full does not, the first stage H_mycelic_full lost them at:

stage it died at	patterns	share of the gap	of which rare
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	1	1%	1
inside the question budget	36	47%	24
candidate formed on right entity + chain	7	9%	3
matched primary rule at ANY confidence	7	9%	3
matched, but confidence < 0.5	3	4%	0
cut from the register (matched at >= 0.5, out-ranked)	23	30%	7

Decomposing the gap to the centralised twin (same algorithm)

Paired on 200 (seed, pattern) pairs at 10,000 users: B4_central_triage reports 74 (37.0%), H_mycelic_full reports 75 (37.5%). B4_central_triage finds 34 that H_mycelic_full misses; H_mycelic_full finds 35 that B4_central_triage misses.

Of the 34 patterns **B4_central_triage** reports and **H_mycelic_full** does not, the first stage **H_mycelic_full** lost them at:

stage it died at	patterns	share of the gap	of which rare
inside the question budget	16	47%	6
candidate formed on right entity + chain	4	12%	1
matched primary rule at ANY confidence	3	9%	1
matched, but confidence < 0.5	3	9%	1
cut from the register (matched at >= 0.5, out-ranked)	8	24%	1

Decomposing the gap to the perfect-retrieval oracle

Paired on 200 (seed, pattern) pairs at 10,000 users: **Y_oracle_retrieval** reports 48 (24.0%), **H_mycelic_full** reports 75 (37.5%). **Y_oracle_retrieval** finds 16 that **H_mycelic_full** misses; **H_mycelic_full** finds 43 that **Y_oracle_retrieval** misses.

Of the 16 patterns **Y_oracle_retrieval** reports and **H_mycelic_full** does not, the first stage **H_mycelic_full** lost them at:

stage it died at	patterns	share of the gap	of which rare
inside the question budget	10	62%	6
candidate formed on right entity + chain	1	6%	0
matched primary rule at ANY confidence	2	12%	1
matched, but confidence < 0.5	1	6%	0
cut from the register (matched at >= 0.5, out-ranked)	2	12%	0

50,000 users (3 seeds)

Stage survival, all patterns

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (>=2 facets, right entity+predicate)	—	0.960 [0.93, 1.00]	0.960 [0.93, 1.00]	0.960 [0.93, 1.00]
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	—	—	0.780 [0.74, 0.80]	—
on the triage candidate list	—	—	0.780 [0.74, 0.80]	—
inside the question budget	—	—	0.427 [0.36, 0.47]	—
descent reached a facet holder	—	—	0.427 [0.35, 0.48]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.493 [0.47, 0.51]	0.450 [0.38, 0.49]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.890 [0.87, 0.91]	0.380 [0.34, 0.41]	0.397 [0.34, 0.44]	0.917 [0.87, 0.97]
matched primary rule at ANY confidence	0.853 [0.84, 0.87]	0.323 [0.31, 0.34]	0.363 [0.31, 0.39]	0.797 [0.74, 0.86]
matched with confidence >= 0.5	0.803 [0.79, 0.82]	0.323 [0.31, 0.34]	0.363 [0.31, 0.39]	0.780 [0.72, 0.84]
survived the register cut (= reported)	0.703 [0.68, 0.74]	0.323 [0.31, 0.34]	0.363 [0.31, 0.39]	0.267 [0.16, 0.34]

patterns counted: A2_chunked_ctx: 300, B4_central_triage: 300, H_mycelic_full: 300, Y_oracle_retrieval: 300

Stage survival, rare-signal patterns only

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (>=2 facets, right entity+predicate)	—	0.905 [0.84, 1.00]	0.905 [0.84, 1.00]	0.905 [0.84, 1.00]
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	—	—	0.558 [0.44, 0.67]	—
on the triage candidate list	—	—	0.558 [0.44, 0.67]	—

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
inside the question budget	—	—	0.307 [0.23, 0.36]	—
descent reached a facet holder	—	—	0.274 [0.15, 0.36]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.213 [0.18, 0.24]	0.314 [0.23, 0.36]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.893 [0.89, 0.90]	0.178 [0.13, 0.25]	0.263 [0.18, 0.32]	0.895 [0.82, 0.96]
matched primary rule at ANY confidence	0.857 [0.85, 0.87]	0.157 [0.10, 0.21]	0.235 [0.15, 0.29]	0.779 [0.67, 0.89]
matched with confidence >= 0.5	0.781 [0.74, 0.82]	0.157 [0.10, 0.21]	0.235 [0.15, 0.29]	0.762 [0.62, 0.89]
survived the register cut (= reported)	0.622 [0.56, 0.68]	0.157 [0.10, 0.21]	0.235 [0.15, 0.29]	0.230 [0.11, 0.32]

patterns counted: A2_chunked_ctx: 112, B4_central_triage: 112, H_mycelic_full: 112, Y_oracle_retrieval: 112

Stage survival, common patterns only

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (>=2 facets, right entity+predicate)	—	1.000 [1.00, 1.00]	1.000 [1.00, 1.00]	1.000 [1.00, 1.00]
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	—	—	0.911 [0.89, 0.93]	—
on the triage candidate list	—	—	0.911 [0.89, 0.93]	—
inside the question budget	—	—	0.501 [0.44, 0.55]	—
descent reached a facet holder	—	—	0.522 [0.48, 0.56]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.662 [0.61, 0.72]	0.533 [0.48, 0.58]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.885 [0.85, 0.92]	0.509 [0.44, 0.59]	0.479 [0.44, 0.51]	0.933 [0.91, 0.97]
matched primary rule at ANY confidence	0.849 [0.82, 0.88]	0.430 [0.36, 0.49]	0.444 [0.41, 0.49]	0.811 [0.79, 0.85]
matched with confidence >= 0.5	0.819 [0.80, 0.84]	0.430 [0.36, 0.49]	0.444 [0.41, 0.49]	0.796 [0.78, 0.82]
survived the register cut (= reported)	0.754 [0.75, 0.76]	0.430 [0.36, 0.49]	0.444 [0.41, 0.49]	0.292 [0.20, 0.35]

patterns counted: A2_chunked_ctx: 188, B4_central_triage: 188, H_mycelic_full: 188, Y_oracle_retrieval: 188

Where the hierarchy loses each pattern (terminal stage)

stage the pattern died at	all	rare	common
extracted (>=2 facets, right entity+predicate)	5 (2%)	5	0
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	60 (20%)	43	17
on the triage candidate list	0 (0%)	0	0
inside the question budget	100 (33%)	29	71
descent reached a facet holder	0 (0%)	0	0
>=2 gold links in kernel pool (= evidence coverage)	0 (0%)	0	0
candidate formed on right entity + chain	15 (5%)	6	9
matched primary rule at ANY confidence	11 (4%)	3	8
matched, but confidence < 0.5	0 (0%)	0	0
cut from the register (matched at >= 0.5, out-ranked)	0 (0%)	0	0
reported	109 (36%)	26	83
total	300	112	188

Why the sketch could not see the invisible ones

why the sketch could not see it	all	rare	common	median witnesses
fewer than 2 foreign sites	15	10	5	5

why the sketch could not see it	all	rare	common	median witnesses
foreign sites in fewer than 2 regions	8	7	1	7
causal span < 2 across foreign sites	43	32	11	7

Discovery by number of witnesses (hierarchy)

witnesses (records)	n	visible to sketch	in pool	reported
0–4	16	0.56	0.44	0.31
5–7	75	0.55	0.28	0.20
8–11	22	0.59	0.32	0.27
12–16	25	0.80	0.32	0.20
17–25	104	0.91	0.49	0.41
26–∞	58	0.97	0.71	0.60

Discovery by number of witnesses (A2_chunked_ctx)

witnesses (records)	n	visible to sketch	in pool	reported
0–4	16	—	1.00	0.44
5–7	75	—	1.00	0.61
8–11	22	—	1.00	0.77
12–16	25	—	1.00	0.84
17–25	104	—	1.00	0.70
26–∞	58	—	1.00	0.81

Where the reported patterns sit in the register

architecture	reported	median rank (0-based, confidence-sorted register)	within top 40	within top 100	median confidence
A2_chunked_ctx	211	851	17	34	0.85
B4_central_triage	97	667	3	9	0.84
H_mycelic_full	109	886	1	5	0.88
Y_oracle_retrieval	80	1268	1	6	0.92

Decomposing the gap to the strongest centralised system

Paired on 300 (seed, pattern) pairs at 50,000 users: A2_chunked_ctx reports 211 (70.3%), H_mycelic_full reports 109 (36.3%). A2_chunked_ctx finds 125 that H_mycelic_full misses; H_mycelic_full finds 23 that A2_chunked_ctx misses.

Of the 125 patterns A2_chunked_ctx reports and H_mycelic_full does not, the first stage H_mycelic_full lost them at:

stage it died at	patterns	share of the gap	of which rare
extracted (>=2 facets, right entity+predicate)	1	1%	1
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	43	34%	29
inside the question budget	67	54%	18
candidate formed on right entity + chain	7	6%	2
matched primary rule at ANY confidence	7	6%	1

Decomposing the gap to the centralised twin (same algorithm)

Paired on 300 (seed, pattern) pairs at 50,000 users: B4_central_triage reports 97 (32.3%), H_mycelic_full reports 109 (36.3%). B4_central_triage finds 32 that H_mycelic_full misses; H_mycelic_full finds 44 that B4_central_triage misses.

Of the 32 patterns **B4_central_triage** reports and **H_mycelic_full** does not, the first stage **H_mycelic_full** lost them at:

stage it died at	patterns	share of the gap	of which rare
inside the question budget	26	81%	1
candidate formed on right entity + chain	2	6%	0
matched primary rule at ANY confidence	4	12%	1

Decomposing the gap to the perfect-retrieval oracle

Paired on 300 (seed, pattern) pairs at 50,000 users: **Y_oracle_retrieval** reports 80 (26.7%), **H_mycelic_full** reports 109 (36.3%). **Y_oracle_retrieval** finds 46 that **H_mycelic_full** misses; **H_mycelic_full** finds 75 that **Y_oracle_retrieval** misses.

Of the 46 patterns **Y_oracle_retrieval** reports and **H_mycelic_full** does not, the first stage **H_mycelic_full** lost them at:

stage it died at	patterns	share of the gap	of which rare
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	17	37%	11
inside the question budget	21	46%	3
candidate formed on right entity + chain	3	7%	0
matched primary rule at ANY confidence	5	11%	1

3. Paired decomposition on identical worlds

Each row is the calibrated hierarchy with one setting changed, run on the same (scale, seed) worlds as the base. **An unbounded register is a benchmark change, not a design**; it appears here only to measure how much discovery is being lost to ranking rather than to retrieval or judgement. Arrows mark a 95% bootstrap interval entirely on one side of zero.

10,000 users

variant	found	evidence cov.	rare recall	AP	FDR	compute	calls
H_mycelic_full (calibrated)	0.375	0.690	0.215	0.0227	0.975	1.09e+06	9.10e+04
ask every triage candidate (question_frac = 1.0) (5 seeds)	0.300	0.985 ↑	0.271	0.0113 ↓	0.980	2.14e+06 ↑	2.21e+05 ↑
unbounded register (diagnostic only) (5 seeds)	0.540 ↑	0.690	0.341 ↑	0.0261 ↑	0.985 ↑	1.16e+06 ↑	9.10e+04
both (5 seeds)	0.795 ↑	0.985 ↑	0.697 ↑	0.0193	0.989 ↑	2.36e+06 ↑	2.21e+05 ↑
full budget, home-site evidence allowed (5 seeds)	0.360	0.985 ↑	0.274	0.0209	0.976	2.76e+06 ↑	2.56e+05 ↑

50,000 users

variant	found	evidence cov.	rare recall	AP	FDR	compute	calls
H_mycelic_full (calibrated)	0.363	0.450	0.235	0.0081	0.987	2.97e+06	2.89e+05
ask every triage candidate (question_frac = 1.0) (3 seeds)	0.407 ↑	0.787 ↑	0.248	0.0085	0.986 ↓	6.27e+06 ↑	6.99e+05 ↑
unbounded register (diagnostic only) (3 seeds)	0.363	0.450	0.235	0.0081	0.987	2.98e+06 ↑	2.89e+05
both (3 seeds)	0.643 ↑	0.787 ↑	0.417 ↑	0.0112 ↑	0.993 ↑	6.83e+06 ↑	6.99e+05 ↑

Reading it. Asking every triage candidate raises evidence coverage from 0.69 to 0.985 at 10k (0.45 to 0.79 at 50k) and *lowers* discovery at 10k: the recovered evidence produces candidates the register cannot hold. Remove the register pressure and the same runs report 0.795 at 10k — above A2's 0.685 — and 0.643 at 50k against A2's 0.703, at two-thirds of A2's compute. Allowing home-site evidence back in does not help, so the flood is not the foreign-only filter. The residual at 50k is the sketch loss, which no question budget can recover because the anchor never reaches the triage list.

4. Why the hierarchy's ranker is worse than A2's on the same operator

Both systems rank with the same confidence logistic. Its main terms — a causal span, several regions, links dominated by different branches — are exactly the properties the triage *pre-selects* anchors for. Among the candidates the hierarchy's kernel actually has to order, those terms carry almost no information, because every questioned anchor already has them. A2's pool holds every anchor, including thousands with no cross-organisational structure at all, for which the same terms are decisive. The hierarchy's discrimination problem is a conditioning problem, not an operator problem.

The consequence is testable: features the triage did **not** condition on — per-link independent support, inter-link lags, conflict volume, dispersion, whether the candidate came from a question — should separate genuine from spurious candidates among the triage survivors. Section 6 measures that.

5. Two smaller losses, measured

Extraction (10% at 50k, 24% of rare patterns). This is stochastic recall, not a capability ceiling: a fresh edge-tier re-read of the same records recovers 80–91% of the patterns lost there, a kernel-tier *local* re-read 100%. A user agent asked a targeted question about an entity can re-read its own notes about it — the entity name is in every record's surface text — so the loss is recoverable without moving any text.

Sketch (18% at 50k, 39 of 54 rare). 43 of the 54 fail on "causal span < 2 across foreign sites": a facet with one witness at a site sets no bit under `sketch_min_support = 2`. Lowering the threshold lets benign single mentions set bits and lengthens the triage list, which only pays if the ranker can absorb the noise — which is the same lever as section 4.

6. How separable are genuine candidates, on kernel-side features?

Every candidate the synthesis produced (before the register cut) was dumped with its confidence-logistic inputs and labelled against gold. One shared logistic was fitted on the pooled candidates of all four systems on the calibration seeds (500–502) and evaluated on the evaluation seeds (0–4) by re-ranking each run's candidates under its own register cap. No feature reads raw text; the privacy accounting is unchanged.

(calibrator evaluation failed: too many indices for array: array is 1-dimensional, but 2 were indexed)

7. What this changed about the next architecture

Not "richer sketches". In order of the loss they addressed, with the measured outcome (section 9 has every paired table):

1. **A calibrated ranker** over kernel-side features, fitted on the calibration seeds by the same protocol as every other knob, so the question budget could be spent without flooding the register. Done: +0.21 found at 10k, +0.16 at 50k, together with the budget it unlocked.
2. **A cheaper descent**, so the budget is affordable at 50k. Done as batched metering (calls –61% to –71% at identical decisions); the chain-scoped variant that halves compute is kept as a separate arm because it costs evidence coverage.
3. **Link timing in the kernel's temporal check**, which the loss replay found contaminated by stale mentions. Done (hybrid): +0.04 at 10k, +0.06 at 50k, free.
4. **Targeted local re-extraction** on descent. Measured and rejected: no gain at 10k, inconclusive at 50k.
5. **A support-1 sketch bit** for rare facets. Measured on one calibration seed and rejected as implemented: it floods the triage list.

Each was a paired experiment against `H_mycelic_full` on identical worlds with the centralised controls re-run at the same time. None moves raw text.

8. Reproducing this

```
python3 -m research.mycelic.loss_accounting      # 10k x 5, 50k x 3 -> loss_funnel.jsonl
python3 -m research.mycelic.loss_report          # the tables
python3 -m research.mycelic.quick_paired --tag diag_qf1 --cfg '{"question_frac": 1.0}' --seeds 0,1,2,3,4
python3 -m research.mycelic.calibrator dump && python3 -m research.mycelic.calibrator fit
python3 -m research.mycelic.loss_doc            # this document
python3 -m research.mycelic.make_pdf docs/MYCELIC_LOSS_ACCOUNTING.md docs/MYCELIC_LOSS_ACCOUNTING.pdf
```

Generated 2026-09-22 by `research/mycelic/loss_doc.py` from the raw artifacts.

9. vNext, measured

Everything in this section is a **paired** experiment: the variant and its base run on identical worlds, and the Δ shown under each variant value is the paired mean with an arrow when its 95% bootstrap interval excludes zero. Evaluation seeds are 0–4 at 10,000 users and 0–2 at 50,000; every knob and every ranker was fitted on the calibration seeds (500–502) before the evaluation seeds were read, and the two tables that use calibration seeds say so. Nothing here changes the register cap, the threshold, the gold labels, the worlds or the matching rule; the unbounded register appears only as a diagnostic.

9.1 A ranker fitted on kernel-side features

The confidence logistic is replaced as a *ranker* only. Version 1 kept the hand-set ≥ 0.5 gate and re-ordered inside it; version 2 keeps the hand-gated **count** and lets the learned score choose which candidates fill those slots, with anchor-level context features; version 3 adds evidence-shape features (origin users, single-witness fraction, echo ratio, lag dispersion), the triage context of the anchor and the kernel's own attribution verdict — 45 kernel-held quantities, no raw text. l2 and the a-priori interaction set are chosen leave-one-seed-out on the calibration seeds; the depth-2 boosted trees lost to the logistic in every selection. Adoption is decided per architecture on the calibration seeds (each system keeps whichever ranker is not worse for it), so no control is compared at a ranker chosen for somebody else.

Version 1 — order within the hand gate (base = same system, hand-set ranker)

experiment	seed s	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
hierarchy, calibrated budget: hand ranker	5	0.375	0.215	0.690	0.0227	0.975	0.180	1.09e+06	9.10e+04
hierarchy, calibrated budget: variant	5	0.440 ^{+0.065} ↑	0.167 ^{-0.048}	0.685 ^{-0.005}	0.1318 ^{+0.1091} ↑	0.970 ^{-0.004} ↓	0.245 ^{+0.065} ↑	1.08e+06 ^{-6.10e+03} ↓	9.08e+04 ^{-2.47e+02}
hierarchy, full question budget: hand ranker	5	0.375	0.215	0.690	0.0227	0.975	0.180	1.09e+06	9.10e+04
hierarchy, full question budget: variant	5	0.450 ^{+0.075} ↑	0.197 ^{-0.018}	0.980 ^{+0.290} ↑	0.0573 ^{+0.0346} ↑	0.970 ^{-0.005} ↓	0.240 ^{+0.060} ↑	2.17e+06 ^{+1.08e+06} ↑	2.26e+05 ^{+1.35e+05} ↑
A2_chunked_ctx: hand ranker	5	0.685	0.653	1.000	0.0669	0.954	0.345	2.08e+06	3.00e+00
A2_chunked_ctx: variant	5	0.650 ^{-0.035} ↓	0.561 ^{-0.092}	1.000 ^{+0.000}	0.1831 ^{+0.1161} ↑	0.957 ^{+0.003} ↑	0.350 ^{+0.005}	2.08e+06 ^{+0.00e+00}	3.00e+00 ^{+0.00e+00}
B4_central_triage: hand ranker	5	0.370	0.258	0.970	0.0219	0.975	0.175	3.98e+05	1.00e+04
B4_central_triage: variant	5	0.505 ^{+0.135} ↑	0.238 ^{-0.020}	0.970 ^{+0.000}	0.1192 ^{+0.0972} ↑	0.966 ^{-0.009} ↓	0.240 ^{+0.065} ↑	3.98e+05 ^{+0.00e+00}	1.00e+04 ^{+0.00e+00}
Y_oracle_retrieval: hand ranker	5	0.240	0.143	1.000	0.0061	0.984	0.125	4.87e+05	1.00e+04
Y_oracle_retrieval: variant	5	0.360 ^{+0.120} ↑	0.120 ^{-0.023}	1.000 ^{+0.000}	0.0232 ^{+0.0171} ↑	0.976 ^{-0.008} ↓	0.200 ^{+0.075} ↑	4.87e+05 ^{+0.00e+00}	1.00e+04 ^{+0.00e+00}

Version 2 — learned top-K gate + anchor context

experiment	seed s	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
hierarchy, calibrated budget: hand ranker	5	0.375	0.215	0.690	0.0227	0.975	0.180	1.09e+06	9.10e+04
hierarchy, calibrated budget: variant	5	0.480 ^{+0.105} ↑	0.253 ^{+0.038}	0.690 ^{+0.000}	0.1334 ^{+0.1107} ↑	0.968 ^{-0.007} ↓	0.275 ^{+0.095} ↑	1.08e+06 ^{-1.17e+04} ↓	9.07e+04 ^{-3.22e+02} ↓
A2_chunked_ctx: hand ranker	5	0.685	0.653	1.000	0.0669	0.954	0.345	2.08e+06	3.00e+00
A2_chunked_ctx: variant	5	0.645 ^{-0.040} ↓	0.566 ^{-0.087} ↓	1.000 ^{+0.000}	0.1999 ^{+0.1329} ↑	0.957 ^{+0.003} ↑	0.380 ^{+0.035}	2.08e+06 ^{+0.00e+00}	3.00e+00 ^{+0.00e+00}
B4_central_triage: hand ranker	5	0.370	0.258	0.970	0.0219	0.975	0.175	3.98e+05	1.00e+04
B4_central_triage: variant	5	0.540 ^{+0.170} ↑	0.280 ^{+0.022}	0.970 ^{+0.000}	0.1298 ^{+0.1078} ↑	0.964 ^{-0.011} ↓	0.295 ^{+0.120} ↑	3.98e+05 ^{+0.00e+00}	1.00e+04 ^{+0.00e+00}
Y_oracle_retrieval: hand ranker	5	0.240	0.143	1.000	0.0061	0.984	0.125	4.87e+05	1.00e+04
Y_oracle_retrieval: variant	5	0.360 ^{+0.120} ↑	0.104 ^{-0.040}	1.000 ^{+0.000}	0.0210 ^{+0.0150} ↑	0.976 ^{-0.008} ↓	0.230 ^{+0.105} ↑	4.87e+05 ^{+0.00e+00}	1.00e+04 ^{+0.00e+00}

Version 3 — evidence shape + triage context + attribution (the hierarchy row also carries the budget and batching changes of 9.2; the controls run at their own settings)

experiment	seed s	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
hierarchy: ranker v3 + question_frac 0.65 + batched descent: hand ranker	5	0.375	0.215	0.690	0.0227	0.975	0.180	1.09e+06	9.10e+04
hierarchy: ranker v3 + question_frac 0.65 + batched descent: variant	5	0.585 ^{+0.210 ↑}	0.390 ^{+0.175 ↑}	0.970 ^{+0.280 ↑}	0.1570 ^{+0.1343 ↑}	0.961 ^{-0.014 ↓}	0.330 ^{+0.150 ↑}	1.36e+06 ^{+2.73e+05 ↑}	2.66e+04 ^{-6.45e+04 ↓}
A2_chunked_ctx: hand ranker	5	0.685	0.653	1.000	0.0669	0.954	0.345	2.08e+06	3.00e+00
A2_chunked_ctx: variant	5	0.745 ^{+0.060 ↑}	0.649 ^{-0.003}	1.000 ^{+0.000}	0.2117 ^{+0.1448 ↑}	0.950 ^{-0.004}	0.445 ^{+0.100 ↑}	2.08e+06 ^{+0.00e+00}	3.00e+00 ^{+0.00e+00}
B4_central_triage: hand ranker	5	0.370	0.258	0.970	0.0219	0.975	0.175	3.98e+05	1.00e+04
B4_central_triage: variant	5	0.560 ^{+0.190 ↑}	0.308 ^{+0.050}	0.970 ^{+0.000}	0.1457 ^{+0.1238 ↑}	0.963 ^{-0.012 ↓}	0.350 ^{+0.175 ↑}	3.98e+05 ^{+0.00e+00}	1.00e+04 ^{+0.00e+00}
Y_oracle_retrieval: hand ranker	5	0.240	0.143	1.000	0.0061	0.984	0.125	4.87e+05	1.00e+04
Y_oracle_retrieval: variant	5	0.425 ^{+0.185 ↑}	0.134 ^{-0.010}	1.000 ^{+0.000}	0.0252 ^{+0.0191 ↑}	0.972 ^{-0.012 ↓}	0.265 ^{+0.140 ↑}	4.87e+05 ^{+0.00e+00}	1.00e+04 ^{+0.00e+00}

What it says. Version 3 lifts the hierarchy by 21 points of discovery on 5/5 seeds, rare recall from 0.21 to 0.39 and evidence coverage to 0.97, and now helps A2_chunked_ctx too (0.685 → 0.745), so every control adopts it. Decoy acceptance rises with the ranker on every system by about +0.15: the planted traps share the features that make a genuine chain look genuine. FDR falls slightly at the same time.

The decoy regression, attacked and not fixed. Up-weighting decoy rows in the fit and penalising decoys in the selection objective chose weight 10 on the calibration seeds; on the evaluation seeds it lost found and rare recall for a 0.05 reduction in decoy acceptance. Rejected; decoy weighting stays available as an explicit option.

arm	seed s	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
ranker v3 (base)	5	0.585	0.390	0.970	0.1570	0.961	0.330	1.36e+06	2.66e+04
decoy-weighted selection (weight 10)	5	0.525 ^{-0.060 ↓}	0.272 ^{-0.118 ↓}	0.970 ^{+0.000}	0.1850 ^{+0.0279}	0.965 ^{+0.004}	0.280 ^{-0.050 ↓}	1.36e+06 ^{-4.70e+03}	2.66e+04 ^{-8.60e+00}

9.2 The question budget, with the ranker fixed

Base = the hierarchy with ranker v3 and batched descent at question_frac 0.50; the v1 calibration had chosen 0.25 because the hand-set confidence let extra candidates flood the register.

arm	seeds	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
question_frac 0.50 (base)	5	0.555	0.356	0.875	0.1572	0.963	0.320	1.20e+06	2.62e+04
question_frac 0.65	5	0.585 ^{+0.030}	0.390 ^{+0.033}	0.970 ^{+0.095 ↑}	0.1570 ^{-0.0002}	0.961 ^{-0.002}	0.330 ^{+0.010}	1.36e+06 ^{+1.58e+05 ↑}	2.66e+04 ^{+3.73e+02 ↑}
question_frac 0.80	5	0.565 ^{+0.010}	0.368 ^{+0.012}	0.985 ^{+0.110 ↑}	0.0537 ^{-0.1035 ↓}	0.962 ^{-0.001}	0.345 ^{+0.025 ↑}	1.54e+06 ^{+3.41e+05 ↑}	2.67e+04 ^{+5.25e+02 ↑}
question_frac 1.00	5	0.565 ^{+0.010}	0.384 ^{+0.028}	0.985 ^{+0.110 ↑}	0.0464 ^{-0.1108 ↓}	0.962 ^{-0.001}	0.315 ^{-0.005}	1.79e+06 ^{+5.84e+05 ↑}	2.68e+04 ^{+5.59e+02 ↑}

What the sweep says. Coverage saturates at 0.65 (0.97); 0.80 and 1.00 buy no discovery, cost +13% and +31% compute and lose AP, because the wider budget floods the 600-entry register with more same-anchor chains than the ranker was fitted to demote. 0.65 is the budget at both scales (9.6).

9.3 Batched metering, and the merge that was rejected

Batching meters one routing call per node and one read per queried user per round, with identical decisions and identical per-record tokens; it is why calls fall by about two thirds while compute rises with the budget. Profiled on one calibration seed (500) at the full budget, single runs, **not** evaluation-seed measurements:

stage	share of compute	calls	note
kernel re-read of the merged pool	42%	1	89,525 objects × 26 tokens in one frontier call
descent routing	31%	145,823	one frontier-tier call per (parent node, anchor)
user reads	6%	69,549	8,361 of 9,491 reached users queried more than once
edge extraction	6%	9,999	unchanged by any of this

change (same seed, full budget, ranker off)	found	AP	rare	pool	compute	calls	kept?
none	0.400	0.0123	0.125	89,525	2.31e+06	227,524	—
batched routing + reads	0.400	0.0123	0.125	89,525	1.91e+06	27,127	yes — pure cost
+ merge descent returns per (predicate, entity) before the kernel reads them	0.275	0.0052	—	22,225	1.07e+06	27,128	no
+ merge per (predicate, entity, polarity)	0.325	0.0075	0.312	23,210	1.08e+06	27,128	no — needs the ranker re-fitted on merged candidates; queued

9.4 Targeted local re-extraction: rejected

When a descent reaches a user, that user re-reads its *own* notes that name the entity but produced no claim on the first pass; raw text stays on the node. One calibration seed promised +0.125. Paired on the evaluation seeds (base = the v1 hierarchy; compare the two variant rows):

experiment	seed s	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
ranker v2, qf 0.65, batched: v1 hierarchy	5	0.480	0.253	0.690	0.1334	0.968	0.275	1.08e+06	9.07e+04
ranker v2, qf 0.65, batched: variant	5	0.520 _{+0.040}	0.318 _{+0.065}	0.970 _{+0.280} ↑	0.1336 _{+0.0002}	0.965 _{-0.003}	0.300 _{+0.025}	1.35e+06 _{+2.75e+05} ↑	2.66e+04 _{-6.42e+04} ↓
same + local re-extraction: v1 hierarchy	5	0.480	0.253	0.690	0.1334	0.968	0.275	1.08e+06	9.07e+04
same + local re-extraction: variant	5	0.515 _{+0.035}	0.277 _{+0.023}	0.970 _{+0.280} ↑	0.1490 _{+0.0156}	0.966 _{-0.002}	0.280 _{+0.005}	1.37e+06 _{+2.93e+05} ↑	2.66e+04 _{-6.41e+04} ↓

No gain at 10k, +0.04/−0.01 on two 50k seeds (9.6), six times the simulator wall time. Off in the frozen configuration.

9.5 Link timing in the temporal check

The temporal DP dated every link at the earliest mention of its (predicate, entity) pair. A descent that returns every mention of an entity drags that date to a stale or routine mention, and the chain-order test then fails: the panel's replay found 20–22 of 25 in-pool gold patterns per 10k seed with at least one link dated that way. Two repairs, both free (no calls, no new candidates): *modal* dates a link at its heaviest witness cluster; *hybrid* does that for links with ≥ 2 agreeing witnesses and lets single-witness links float across their clusters as separate DP states.

Screen on the calibration seeds 500–502 (ranker v3 not refitted; these are NOT evaluation-seed numbers):

arm	seed s	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
qf 0.65 + batched, min timing (base)	3	0.542	0.187	0.925	0.2112	0.964	0.467	1.39e+06	2.67e+04
modal timing	3	0.583 _{+0.042}	0.232 _{+0.045}	0.917 _{-0.008}	0.2375 _{+0.0264}	0.961 _{-0.003} ↓	0.375 _{-0.092} ↓	1.39e+06 _{-3.60e+03} ↓	2.67e+04 _{-6.13e+01} ↓
hybrid timing	3	0.625 _{+0.083}	0.367 _{+0.180}	0.917 _{-0.008}	0.1939 _{-0.0173}	0.958 _{-0.006}	0.367 _{-0.100} ↓	1.37e+06 _{-2.85e+04} ↓	2.66e+04 _{-1.53e+02} ↓
chain-scoped questions (strict)	3	0.550 _{+0.008}	0.121 _{-0.066}	0.750 _{-0.175} ↓	0.2177 _{+0.0065}	0.963 _{-0.001}	0.458 _{-0.008}	7.29e+05 _{-6.66e+05} ↓	2.64e+04 _{-2.97e+02} ↓
chain-scoped + modal	3	0.583 _{+0.042}	0.197 _{+0.010}	0.750 _{-0.175} ↓	0.2337 _{+0.0225} ↑	0.961 _{-0.003} ↓	0.367 _{-0.100} ↓	7.28e+05 _{-6.67e+05} ↓	2.64e+04 _{-3.47e+02} ↓
chain-scoped + hybrid	3	0.600 _{+0.058}	0.285 _{+0.098}	0.758 _{-0.167} ↓	0.2211 _{+0.0099}	0.960 _{-0.004}	0.425 _{-0.042}	7.31e+05 _{-6.64e+05} ↓	2.63e+04 _{-4.13e+02} ↓

Evaluation seeds 0–4, ranker refitted on the calibration seeds under each pipeline:

arm	seed s	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
min timing, ranker v3 (base)	5	0.585	0.390	0.970	0.1570	0.961	0.330	1.36e+06	2.66e+04

arm	seed s	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
hybrid, ranker v3 (not refitted)	5	0.630 _{+0.045}	0.415 _{+0.025}	0.980 _{+0.010}	0.1734 _{+0.0164}	0.958 _{-0.003}	0.320 _{-0.010}	1.36e+06 _{-5.70e+03}	2.66e+04 _{+6.20e+00}
hybrid, refitted ranker	5	0.625 _{+0.040}	0.375 _{-0.015}	0.980 _{+0.010}	0.1849 _{+0.0279}	0.958 _{-0.003}	0.360 _{+0.030}	1.37e+06 _{+7.28e+03}	2.67e+04 _{+1.34e+02}
hybrid + chain-scoped questions (lean)	5	0.610 _{+0.025}	0.318 _{-0.071}	0.800 _{-0.170} ↓	0.1944 _{+0.0373}	0.959 _{-0.002}	0.375 _{+0.045}	7.26e+05 _{-6.35e+05} ↓	2.64e+04 _{-1.81e+02}

arm	seeds	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
min timing, ranker v3 (base)	5	0.585	0.390	0.970	0.1570	0.961	0.330	1.36e+06	2.66e+04
modal, ranker v3 (not refitted)	5	0.550 _{-0.035}	0.290 _{-0.100}	0.970 _{+0.000}	0.1710 _{+0.0140}	0.963 _{+0.002}	0.340 _{+0.010}	1.36e+06 _{-4.13e+02}	2.67e+04 _{+7.38e+01}
modal, refitted ranker	5	0.575 _{-0.010}	0.332 _{-0.058}	0.965 _{-0.005}	0.1748 _{+0.0178}	0.962 _{+0.001}	0.340 _{+0.010}	1.36e+06 _{+3.48e+03}	2.66e+04 _{-4.00e-01}

Modal did not replicate (+0.04 on the calibration seeds, -0.01 on the evaluation seeds); hybrid did (+0.04, no seed worse). The refitted ranker goes forward because that is the pre-registered procedure; the previous ranker's better rare recall and decoy acceptance under hybrid timing are reported, not acted on, because acting on them would be selection on the held-out seeds. The chain-scoped variant halves compute at a cost in coverage and rare recall and is kept as a separate arm (H_mycelic_lean).

9.6 50,000 users

arm	see ds	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
v1 hierarchy (hand ranker, qf 0.25) (base)	3	0.363	0.235	0.450	0.0081	0.987	0.197	2.97e+06	2.89e+05
ranker v3, qf 0.65, batched	3	0.527 _{+0.163} ↑	0.335 _{+0.100} ↑	0.723 _{+0.273} ↑	0.0775 _{+0.0694} ↑	0.982 _{-0.005} ↓	0.333 _{+0.137} ↑	3.92e+06 _{+9.48e+05} ↑	1.14e+05 _{-1.75e+05} ↓
ranker v3, qf 0.80, batched	3	0.533 _{+0.170} ↑	0.303 _{+0.067} ↑	0.770 _{+0.320} ↑	0.0745 _{+0.0664} ↑	0.982 _{-0.005} ↓	0.367 _{+0.170} ↑	4.39e+06 _{+1.42e+06} ↑	1.16e+05 _{-1.73e+05} ↓
ranker v3, qf 1.00, batched	3	0.533 _{+0.170} ↑	0.294 _{+0.059} ↑	0.787 _{+0.337} ↑	0.0346 _{+0.0265} ↑	0.982 _{-0.005} ↓	0.360 _{+0.163} ↑	5.15e+06 _{+2.18e+06} ↑	1.18e+05 _{-1.72e+05} ↓
ranker v3, qf 0.65 + local re-extraction	3	0.533 _{+0.170} ↑	0.310 _{+0.075} ↑	0.720 _{+0.270} ↑	0.0829 _{+0.0747} ↑	0.982 _{-0.005} ↓	0.373 _{+0.177} ↑	3.94e+06 _{+9.70e+05} ↑	1.14e+05 _{-1.75e+05} ↓
A2_chunked_ctx, hand ranker	3	0.703 _{+0.340} ↑	0.622 _{+0.386} ↑	1.000 _{+0.550} ↑	0.0594 _{+0.0513} ↑	0.976 _{-0.011} ↓	0.373 _{+0.177} ↑	1.02e+07 _{+7.21e+06} ↑	1.00e+01 _{-2.89e+05} ↓
A2_chunked_ctx, ranker v3	3	0.780 _{+0.417} ↑	0.677 _{+0.441} ↑	1.000 _{+0.550} ↑	0.1130 _{+0.1049} ↑	0.974 _{-0.013} ↓	0.443 _{+0.247} ↑	1.02e+07 _{+7.21e+06} ↑	1.00e+01 _{-2.89e+05} ↓

arm	seed s	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
ranker v3, qf 0.65, batched (base)	3	0.527	0.335	0.723	0.0775	0.982	0.333	3.92e+06	1.14e+05
hybrid timing, refitted ranker	3	0.587 _{+0.060} ↑	0.343 _{+0.009}	0.727 _{+0.003}	0.0981 _{+0.0206} ↑	0.980 _{-0.002} ↓	0.373 _{+0.040}	3.91e+06 _{-2.44e+03}	1.14e+05 _{-4.37e+01}
hybrid + chain-scoped questions	3	0.523 _{-0.003}	0.263 _{-0.072} ↓	0.580 _{-0.143} ↓	0.1647 _{+0.0872} ↑	0.982 _{-0.000}	0.340 _{+0.007}	2.71e+06 _{-1.20e+06} ↓	1.13e+05 _{-1.22e+03} ↓

At 50k the register (2,999 slots) is not the binding stage; coverage (0.72) is. The budget saturates at 0.65 here too, re-extraction does not pay, and hybrid timing adds +0.06 on 3/3 seeds at equal compute. The hierarchy ends at about 0.59 against A2's 0.77 with the same ranker, at 38% of A2's compute; that gap is the question budget's reach, not the register.

9.7 The ledger, the cost of each accepted change, and what is frozen

change	scale	seeds	Δ found	Δ rare	Δ cov.	Δ decoy	compute	calls	status
ranker v1: learned order inside the hand gate	10,000	eval 0-4	+0.065 (5/5 better)	-0.048	-0.005	+0.065	-1%	-0%	superseded
ranker v2: learned top-K, anchor-context features	10,000	eval 0-4	+0.105 (5/5 better)	+0.038	+0.000	+0.095	-1%	-0%	superseded
ranker v3 + question_frac 0.65 + batched descent	10,000	eval 0-4	+0.210 (5/5 better)	+0.175	+0.280	+0.150	+25%	-71%	ACCEPTED
decoy-weighted ranker selection	10,000	eval 0-4	-0.060 (1/5 better)	-0.118	+0.000	-0.050	-0%	-0%	rejected
question_frac 0.80 (vs 0.65)	10,000	eval 0-4	-0.020 (1/4 better)	-0.022	+0.015	+0.015	+13%	+1%	rejected
question_frac 1.00 (vs 0.65)	10,000	eval 0-4	-0.020 (1/3 better)	-0.005	+0.015	-0.015	+31%	+1%	rejected
local re-extraction at the user (vs qf 0.65 batched)	10,000	eval 0-4	-0.005 (3/5 better)	-0.042	+0.000	-0.020	+1%	+0%	rejected
batched descent metering alone (ranker v2, qf 0.65; same decisions)	10,000	eval 0-4	+0.000 (0/0 better)	+0.000	+0.000	+0.000	-17%	-84%	ACCEPTED
modal link timing (+ refitted ranker)	10,000	eval 0-4	-0.010 (2/5 better)	-0.058	-0.005	+0.010	+0%	-0%	rejected

change	scale	seeds	Δ found	Δ rare	Δ cov.	Δ decoy	compute	calls	status
hybrid link timing (+ refitted ranker)	10,000	eval 0-4	+0.040 (2/2 better)	-0.015	+0.010	+0.030	+1%	+1%	ACCEPTED
hybrid + chain-scoped questions (H_mycelic_lean)	10,000	eval 0-4	+0.025 (3/4 better)	-0.071	-0.170	+0.045	-47%	-1%	Pareto arm
ranker v3 + qf 0.65 + batched, 50k	50,000	eval 0-2	+0.163 (3/3 better)	+0.100	+0.273	+0.137	+32%	-61%	ACCEPTED
question_frac 0.80 at 50k	50,000	eval 0-2	+0.007 (2/3 better)	-0.032	+0.047	+0.033	+12%	+2%	rejected
local re-extraction at 50k	50,000	eval 0-1	+0.007 (1/3 better)	-0.025	-0.003	+0.040	+1%	-0%	rejected
hybrid link timing at 50k (+ refitted ranker)	50,000	eval 0-2	+0.060 (3/3 better)	+0.009	+0.003	+0.040	-0%	-0%	validation
hybrid + chain-scoped questions at 50k	50,000	eval 0-2	-0.003 (1/3 better)	-0.072	-0.143	+0.007	-31%	-1%	Pareto arm
unbounded register (DIAGNOSTIC ONLY, not a fix)	10,000	eval 0-4	+0.165 (5/5 better)	+0.127	+0.000	+0.150	+7%	+0%	diagnostic
unbounded register + full question budget (DIAGNOSTIC)	10,000	eval 0-4	+0.420 (5/5 better)	+0.482	+0.295	+0.295	+117%	+143%	diagnostic
chain-scoped questions alone (calibration seeds)	10,000	cal 500-502	+0.008 (2/3 better)	-0.066	-0.175	-0.008	-48%	-1%	screen

Compute per accepted change, 10k evaluation seeds:

configuration	found	rare recall	compute	calls	Δ found per +10% compute
v1 hierarchy (hand ranker, qf 0.25)	0.375	0.215	1.09e+06	9.10e+04	—
+ ranker v3, qf 0.65, batched descent	0.585	0.390	1.36e+06	2.66e+04	+0.084
+ hybrid link timing (refitted ranker)	0.625	0.375	1.37e+06	2.67e+04	+0.040 at +0.5% compute

Frozen configuration (`calibration.json` , with the file that justified each value):

knob	frozen value	chosen on	evidence
<code>question_frac</code>	<code>0.65</code>	calibration seeds 500–502, confirmed on held-out 0–4	<code>quick_qf_v3.jsonl</code>
<code>batched_descent</code>	<code>True</code>	calibration seeds 500–502, confirmed on held-out 0–4	<code>quick_v3_H.jsonl</code>
<code>link_time</code>	<code>hybrid</code>	calibration seeds 500–502, confirmed on held-out 0–4	<code>quick_refit_hyb.jsonl</code>
<code>local_reextract</code>	<code>False</code>	calibration seeds 500–502, confirmed on held-out 0–4	<code>quick_rx_reextract.jsonl</code>
<code>strict_targeting</code>	<code>False</code>	calibration seeds 500–502, confirmed on held-out 0–4	<code>quick_cal_screen.jsonl</code>
<code>triage_target_chains</code>	<code>none</code>	calibration seeds 500–502, confirmed on held-out 0–4	<code>quick_cal_screen.jsonl</code>
ranker	logistic, l2 0.3, interactions True, 23614 candidates	seeds [500, 501, 502]	<code>research/mycelic/artifacts/calibration.hyb.json</code>
ranker adopted by	A2_chunked_ctx, B2_map_reduce, B4_central_triage, D_hier_nolineage, E_hier_lineage, F_hier_retrieval, G_hier_questions, H_mycelic_full, I_mycelic_completion, J_mycelic_verified, Y_oracle_retrieval	calibration seeds	<code>ranker_arch_table</code>

The full benchmark is rerun on this configuration by `final_rerun.sh` (every experiment, the controls on identical worlds, the loss accounting at both scales, the candidate dump with an out-of-sample ranker evaluation), and the old-vs-new comparison, the architecture, the experiment matrix and the adversarial review are in `docs/mycelic_vnext/`.